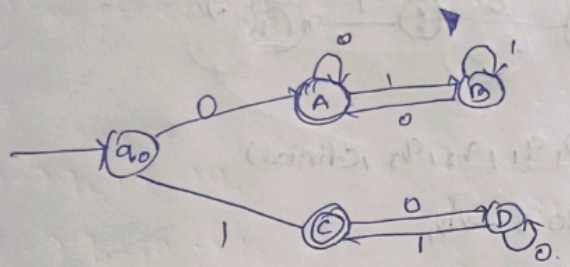
Assignment - 1

① DFA, start and end with the same symbols.



- q_0 = start
- (A) = started 0, end 0 (final)
- B = started 0, end 1
- C = started 1, end 1 (final)
- D = started 1, end 0

state	0	1
$\rightarrow q_0$	A	C
*A	A	B
B	A	B
*C	D	C
D	D	C

Formal definition $M = (Q, \Sigma, \delta, q_0, F)$:

$Q = \{q_0, A, B, C, D\}$

$\Sigma = \{0, 1\}$

q_0 : start state

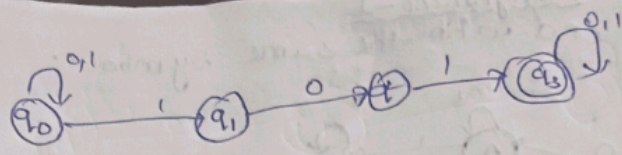
δ = the table above

$F = \{A, C\}$

check 010100:

$q_0 \rightarrow 0 \rightarrow A \rightarrow 1 \rightarrow B \rightarrow 0 \rightarrow A \rightarrow 0 \rightarrow A \rightarrow 0 \leftarrow A \leftarrow 0 \leftarrow B$

- A is final, so accepted.

② 

▼ State: q_0 (start), q_1, q_2, q_3 (finial)

q_0 : 0,1 → q_0 : 1 → q_1

q_1 : 0 → q_2

q_2 : 1 → q_3

q_3 : 0,1 → q_3

State	0	1
→ q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	ϕ
q_2	ϕ	$\{q_3\}$
q_3	$\{q_3\}$	$\{q_3\}$

③ Claim : $P(n) = \Sigma i(i+1) = n(n+1)(n+2)/3$

Base $n = 1$

- LHS $= 1 \cdot 2 = 2$
- RHS $= 1 \cdot 2 \cdot 3/3 = 2$

- Assume $P(K) = \Sigma (i = 1..K) i(i+1) = (K+1)(K+2)/3$

show $P(K+1)$

$\Sigma (i = 1..K+1) = K(K+1)(K+2)/3 + (K+1)(K+2)$

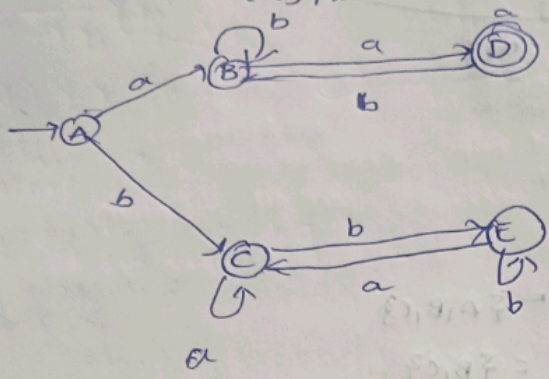
$= (K+1)(K+2)\{K/3+1\}$

$= (K+1)(K+2)(K+3)/3$

• By induction, P(n) is true for all n≥1.

(H) Statement = A = {q0}, B = {q1}, C = {q3}, D = {q1, q2}

E = {q3, q2}



• q_0 : $a \to q_1, b \to q_3$

q_1 : $a, b \to q_1 : a \to q_2$

q_3 : $a, b \to q_3$; $b \to q_2$

• q_2 : final

DFA state

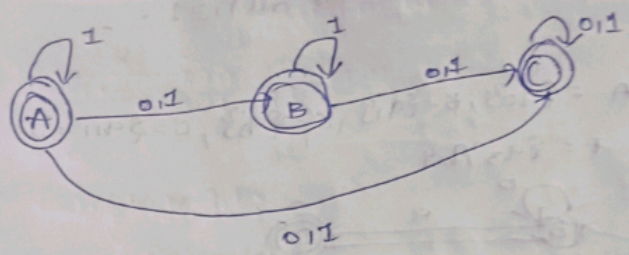
	a	b
→ A = {q0}	B	C
B = {q1}	D	B
C = {q3}	C	E
*D = {q1, q2}	D	B
E = {q3, q2}	C	E

start = A

Final = D, E (they contain q_2)

5 states, no more new subsets appears.

⑤



Given

ε-closures:

ε-Closure (A) = {A, B, C}

ε-closure (B) = {B, C}

ε-closure (C) = {C}

Formula : $\delta_e(q, K) = \varepsilon\text{-Closure}(\delta(\varepsilon\text{-closure}(q), K))$

Computation :

A on 0 : δ (A, B, C}, 0) = {B, C} ∪ ∅ ∪ {C} = {B C}

A on 1 : δ ({A, B, C}, 1) = {A} ∪ {B} ∪ {C} = {A, B, C}

B on 0 : δ ({B, C}, 0) = ∅ ∪ {C} = {C}

B on 1 : δ({B, C}, 1} = {B} ∪ {C} = {B, C}

C on 0 : {C}

C on 1 : {C}

State	0	1
→*A	{B, C}	{A, B, C}
*B	{C}	{B, C}
*C	{C}	{C}

- Intitial state : A (marked)
- Final state : A, B, C (each one's ε-closure contains c)